

Prácticas de Administración de Discos

(Linux + Windows)

PRÁCTICAS EN LINUX

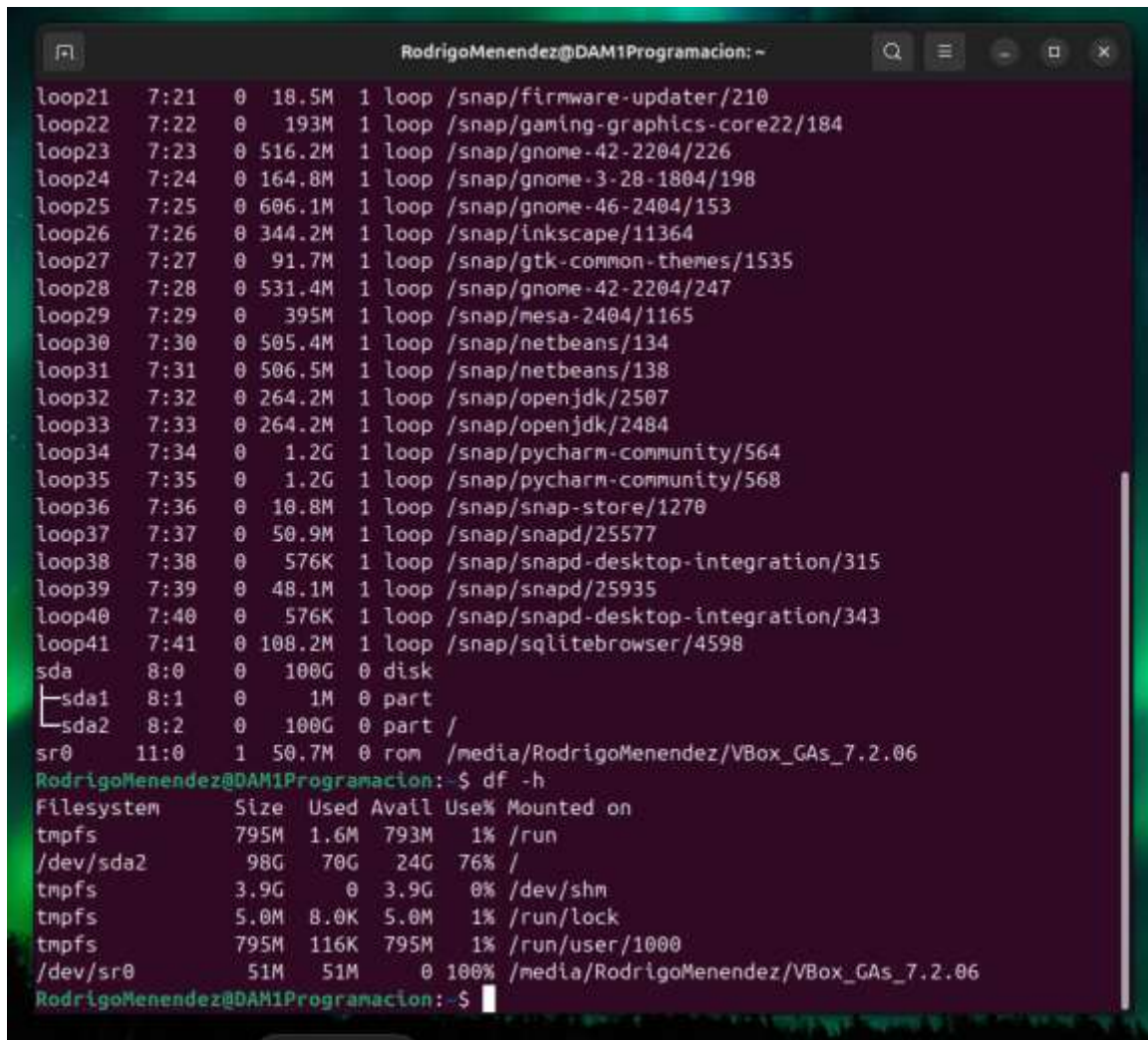
Práctica 1: Ver información básica de discos

Ver todos los discos y particiones:

```
lsblk
```

Ver espacio libre:

```
df -h
```



```
RodrigoMenendez@DAM1Programacion: ~  
loop21 7:21 0 18.5M 1 loop /snap/firmware-updater/210  
loop22 7:22 0 193M 1 loop /snap/gaming-graphics-core22/184  
loop23 7:23 0 516.2M 1 loop /snap/gnome-42-2204/226  
loop24 7:24 0 164.8M 1 loop /snap/gnome-3-28-1804/198  
loop25 7:25 0 606.1M 1 loop /snap/gnome-46-2404/153  
loop26 7:26 0 344.2M 1 loop /snap/inkscape/11364  
loop27 7:27 0 91.7M 1 loop /snap/gtk-common-themes/1535  
loop28 7:28 0 531.4M 1 loop /snap/gnome-42-2204/247  
loop29 7:29 0 395M 1 loop /snap/mesa-2404/1165  
loop30 7:30 0 505.4M 1 loop /snap/netbeans/134  
loop31 7:31 0 506.5M 1 loop /snap/netbeans/138  
loop32 7:32 0 264.2M 1 loop /snap/openjdk/2507  
loop33 7:33 0 264.2M 1 loop /snap/openjdk/2484  
loop34 7:34 0 1.2G 1 loop /snap/pycharm-community/564  
loop35 7:35 0 1.2G 1 loop /snap/pycharm-community/568  
loop36 7:36 0 10.8M 1 loop /snap/snap-store/1270  
loop37 7:37 0 50.9M 1 loop /snap/snapd/25577  
loop38 7:38 0 576K 1 loop /snap/snapd-desktop-integration/315  
loop39 7:39 0 48.1M 1 loop /snap/snapd/25935  
loop40 7:40 0 576K 1 loop /snap/snapd-desktop-integration/343  
loop41 7:41 0 108.2M 1 loop /snap/sqlitebrowser/4598  
sda 8:0 0 100G 0 disk  
├─sda1 8:1 0 1M 0 part  
└─sda2 8:2 0 100G 0 part /  
sr0 11:0 1 50.7M 0 rom /media/RodrigoMenendez/VBox_GAs_7.2.06  
RodrigoMenendez@DAM1Programacion:~$ df -h  
Filesystem      Size  Used Avail Use% Mounted on  
tmpfs            795M  1.6M  793M   1% /run  
/dev/sda2        98G   70G   24G  76% /  
tmpfs            3.9G   0  3.9G   0% /dev/shm  
tmpfs            5.0M   8.0K  5.0M   1% /run/lock  
tmpfs            795M  116K  795M   1% /run/user/1000  
/dev/sr0         51M   51M   0 100% /media/RodrigoMenendez/VBox_GAs_7.2.06  
RodrigoMenendez@DAM1Programacion:~$
```

Práctica 2: Crear y montar una carpeta como unidad

1. Crear carpeta para montar:

```
sudo mkdir /mnt/micarpeta
```

```
sudo mkdir -p /mnt/micarpeta
```

2. Montar una carpeta existente:

```
sudo mount --bind /home/usuario/Documentos /mnt/micarpeta
```

3. Verificar:

```
ls /mnt/micarpeta
```

```

    dmesg(1) may have more information after failed mount system call.
RodrigoMenendez@DAM1Programacion:/$ ls /mnt/micarpeta
RodrigoMenendez@DAM1Programacion:/$ sudo mount --bind /home/Documents/mnt/micarpeta
mount: /home/Documents/mnt/micarpeta: mount point does not exist.
    dmesg(1) may have more information after failed mount system call.
RodrigoMenendez@DAM1Programacion:/$ sudo mount --bind /home/Rodrigo/Documents/mnt/micar
peta
mount: /home/Rodrigo/Documents/mnt/micarpeta: mount point does not exist.
    dmesg(1) may have more information after failed mount system call.
RodrigoMenendez@DAM1Programacion:/$ sudo mount --bind /home/RodrigoMenendez/Documents/n
nt/micarpeta
mount: /home/RodrigoMenendez/Documents/mnt/micarpeta: mount point does not exist.
    dmesg(1) may have more information after failed mount system call.
RodrigoMenendez@DAM1Programacion:/$ whoami
RodrigoMenendez
RodrigoMenendez@DAM1Programacion:/$ sudo mkdir -p /mnt/micarpeta
RodrigoMenendez@DAM1Programacion:/$ ls /home/RodrigoMenendez
composer.phar      IdeaProjects      Public
copiadeseguridad.sql  nongodb-compass_1.46.10_amd64.deb  PycharmProjects
Desktop           Music             snap
Documents         NetBeansProjects Templates
Downloads         Pictures          Videos
eclipse-workspace practica-git-individual
RodrigoMenendez@DAM1Programacion:/$ sudo mount --bind /home/RodrigoMenendez/Documentos
/mnt/micarpeta
mount: /mnt/micarpeta: special device /home/RodrigoMenendez/Documentos does not exist.
    dmesg(1) may have more information after failed mount system call.
RodrigoMenendez@DAM1Programacion:/$ sudo mount --bind /home/RodrigoMenendez/Documents /
mnt/micarpeta
RodrigoMenendez@DAM1Programacion:/$ ls /mnt/micarpeta
'ACII y ANSI.txt'  GitHub  GitHub21-11-25  'GitHub 28-01-2026'
RodrigoMenendez@DAM1Programacion:/$ ls /mnt/micarpeta
'ACII y ANSI.txt'  GitHub  GitHub21-11-25  'GitHub 28-01-2026'
RodrigoMenendez@DAM1Programacion:/$
```

Práctica 3: Ver qué archivos son más grandes

Buscar los 10 archivos más grandes en Documentos:

```
find ~/Documentos -type f -exec du -h {} + | sort -rh | head -10
```

```
RodrigoMenendez@DAM1Programacion: /
RodrigoMenendez@DAM1Programacion:/$ sudo mount --bind /home/RodrigoMenendez/Documentos /mnt/micarpeta
mount: /mnt/micarpeta: special device /home/RodrigoMenendez/Documentos does not exist.
dmesg(1) may have more information after failed mount system call.
RodrigoMenendez@DAM1Programacion:/$ sudo mount --bind /home/RodrigoMenendez/Documentos /mnt/micarpeta
RodrigoMenendez@DAM1Programacion:/$ ls /mnt/micarpeta
'ACII y ANSI.txt'  'GitHub'  'GitHub21-11-25'  'GitHub 28-01-2026'
RodrigoMenendez@DAM1Programacion:/$ ls /mnt/micarpeta
'ACII y ANSI.txt'  'GitHub'  'GitHub21-11-25'  'GitHub 28-01-2026'
RodrigoMenendez@DAM1Programacion:/$ find -/Documents -type f -exec du -h {} + | sort -r h | head -10
576M    /home/RodrigoMenendez/Documents/GitHub 28-01-2026/ProgramacionDAM24-25/Programacion/012 IA/003 Bases de datos vectoriales/chroma_data/chroma.sqlite3
312M    /home/RodrigoMenendez/Documents/GitHub/DAM-24-25/.git/objects/pack/pack-ab6784e1705ae2f3ad39b8138c223d3f44df788b.pack
136M    /home/RodrigoMenendez/Documents/GitHub 28-01-2026/ProgramacionDAM24-25/.git/objects/pack/pack-0553e32cf8c1188da0e71825b4316da8f9332555.pack
134M    /home/RodrigoMenendez/Documents/GitHub21-11-25/ProgramacionDAM24-25/.git/objects/pack/pack-857172018636a8aaf892fe00efe2a2c5a28f0e52.pack
93M     /home/RodrigoMenendez/Documents/GitHub/DAM-24-25/Sistemas informaticos/WindowsServer/DNS, DHCP Y GPO en Active Directory.docx
93M     /home/RodrigoMenendez/Documents/GitHub 28-01-2026/ProgramacionDAM24-25/Sistemas informaticos/WindowsServer/DNS, DHCP Y GPO en Active Directory.docx
92M     /home/RodrigoMenendez/Documents/GitHub 28-01-2026/ProgramacionDAM24-25/.git/objects/93/04a12418f5ed10f3f9df784a2a74d67e5f057f
87M     /home/RodrigoMenendez/Documents/GitHub 28-01-2026/ProgramacionDAM24-25/Programacion/012 IA/003 Bases de datos vectoriales/chroma_data/316f704d-cf46-434c-88bb-4c51fa17f253/data_level0.bin
40M     /home/RodrigoMenendez/Documents/GitHub/DAM-24-25/Programacion/010 Programacion en el lado del servidor/013 Idiomas en progeamacion/a.mp4
40M     /home/RodrigoMenendez/Documents/GitHub 28-01-2026/ProgramacionDAM24-25/Programacion/010 Programacion en el lado del servidor/013 Idiomas en progeamacion/a.mp4
RodrigoMenendez@DAM1Programacion:/$
```

Práctica 4: Ver procesos que usan disco

Ver qué programas están usando disco:

iotop

Si no tienes iotop, instálalo:

```
sudo apt install iotop # Ubuntu/Debian
```


Total DISK READ:		0.00 B/s	Total DISK WRITE:		34.05 K/s
Current DISK READ:		0.00 B/s	Current DISK WRITE:		34.05 K/s
TID	PRI	USER	DISK READ	DISK WRITE	COMMAND
2017	be/4	mongod	0.00 B/s	34.05 K/s	mongod --config /etc/mongod.conf [ftdc]
1	be/4	root	0.00 B/s	0.00 B/s	init splash
2	be/4	root	0.00 B/s	0.00 B/s	[kthreadd]
3	be/4	root	0.00 B/s	0.00 B/s	[pool_workqueue_release]
4	be/0	root	0.00 B/s	0.00 B/s	[kworker/R-rcu_gp]
5	be/0	root	0.00 B/s	0.00 B/s	[kworker/R-sync_wq]
6	be/0	root	0.00 B/s	0.00 B/s	[kworker/R-kvfree_rcu_reclaim]
7	be/0	root	0.00 B/s	0.00 B/s	[kworker/R-slub_flushwq]
8	be/0	root	0.00 B/s	0.00 B/s	[kworker/R-netns]
11	be/0	root	0.00 B/s	0.00 B/s	[kworker/0:0H-events_highpri]
12	be/4	root	0.00 B/s	0.00 B/s	[kworker/u16:0-ipv6_addrconf]
13	be/0	root	0.00 B/s	0.00 B/s	[kworker/R-mm_percpu_wq]
14	be/4	root	0.00 B/s	0.00 B/s	[rcu_tasks_kthread]
15	be/4	root	0.00 B/s	0.00 B/s	[rcu_tasks_rude_kthread]
16	be/4	root	0.00 B/s	0.00 B/s	[rcu_tasks_trace_kthread]
17	be/4	root	0.00 B/s	0.00 B/s	[ksoftirqd/0]
18	be/4	root	0.00 B/s	0.00 B/s	[rcu_preempt]
19	be/4	root	0.00 B/s	0.00 B/s	[rcu_exp_par_gp_kthread_worker/0]
20	be/4	root	0.00 B/s	0.00 B/s	[rcu_exp_gp_kthread_worker]
21	rt/4	root	0.00 B/s	0.00 B/s	[migration/0]
22	rt/4	root	0.00 B/s	0.00 B/s	[idle_inject/0]
23	be/4	root	0.00 B/s	0.00 B/s	[cpuhp/0]
24	be/4	root	0.00 B/s	0.00 B/s	[cpuhp/1]
25	rt/4	root	0.00 B/s	0.00 B/s	[idle_inject/1]
26	rt/4	root	0.00 B/s	0.00 B/s	[migration/1]
27	be/4	root	0.00 B/s	0.00 B/s	[ksoftirqd/1]
29	be/0	root	0.00 B/s	0.00 B/s	[kworker/1:0H-events_highpri]
30	be/4	root	0.00 B/s	0.00 B/s	[cpuhp/2]
31	rt/4	root	0.00 B/s	0.00 B/s	[idle_inject/2]
32	rt/4	root	0.00 B/s	0.00 B/s	[migration/2]
33	be/4	root	0.00 B/s	0.00 B/s	[ksoftirqd/2]
keys: any: refresh q: quit i: ionice o: active p: procs a: accum sort: r: asc left: DISK READ right: COMMAND home: TID end: COMMAND					
CONFIG_TASK_DELAY_ACCT and kernel.task_delayacct sysctl not enabled in kernel, cannot determin					

Práctica 5: Limpiar caché del sistema

Liberar memoria caché de disco (seguro):

```
sudo sync && sudo sysctl -w vm.drop_caches=3
```

Limpiar paquetes temporales:

```
sudo apt autoremove && sudo apt autoclean
```

```
RodrigoMenendez@DAM1Programacion: /
The Linux kernel interfaces that iotop relies on now require root privileges
or the NET_ADMIN capability. This change occurred because a security issue
(CVE-2011-2494) was found that allows leakage of sensitive data across user
boundaries. If you require the ability to run iotop as a non-root user, please
configure sudo to allow you to run iotop as root.

Please do not file bugs on iotop about this.
RodrigoMenendez@DAM1Programacion:/$ sudo iotop
RodrigoMenendez@DAM1Programacion:/$ sudo sync && sudo sysctl -w vm.drop_caches=3
vm.drop_caches = 3
RodrigoMenendez@DAM1Programacion:/$ sudo apt autoremove && sudo apt autoclean
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
The following packages will be REMOVED:
  libllvm19
0 upgraded, 0 newly installed, 1 to remove and 50 not upgraded.
After this operation, 129 MB disk space will be freed.
Do you want to continue? [Y/n] y
(Reading database ... 213086 files and directories currently installed.)
Removing libllvm19:amd64 (1:19.1.1-1ubuntu1~24.04.2) ...
Processing triggers for libc-bin (2.39-0ubuntu8.7) ...
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
RodrigoMenendez@DAM1Programacion:/$
```

PRÁCTICAS EN WINDOWS

Práctica 1: Ver información del disco

1. Abre PowerShell como administrador

2. Escribe:

```
Get-Disk
```

```
Get-Partition
```

```
Get-Volume
```

```
sda 8:0 0 100G 0 disk
Windows PowerShell
Copyright (C) Microsoft Corporation. Todos los derechos reservados.

Prueba la nueva tecnología PowerShell multiplataforma https://aka.ms/pscore6

PS C:\Users\RODRIGOMENENDEZMOLIN> Get-Disk

Number Friendly Name Serial Number HealthStatus OperationalStatus Total Size Partition Style
-----
0 HP SSD EX9... HASE43440600348 _00000001. Healthy Online 953.87 GB GPT
1 Mass Stora... 125D20140310 Healthy No Media 0 B RAW

PS C:\Users\RODRIGOMENENDEZMOLIN> Get-Partition

DiskPath: \\?\scsi#disk&ven_nvme&prod_hp_ssd_ex900_plu#5&114d1f46&0&000000#{53f56307-b6bf-11d0-94f2-00a0c91efb8b}

PartitionNumber DriveLetter Offset Size Type
-----
1 1048576 100 MB System
2 105906176 16 MB Reserved
3 122683392 953.21 GB Basic
4 1023621988352 559 MB Recovery

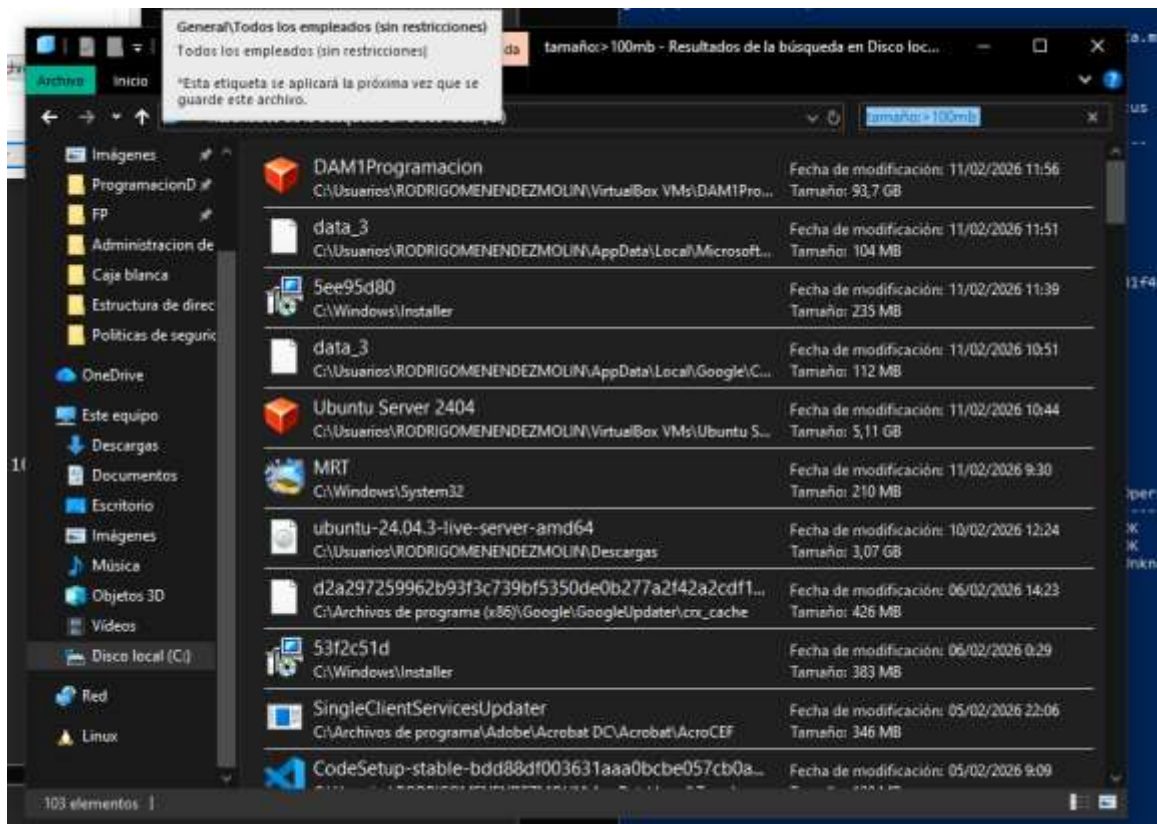
PS C:\Users\RODRIGOMENENDEZMOLIN> Get-Volume

DriveLetter FriendlyName FileSystemType DriveType HealthStatus OperationalStatus SizeRemaining Size
-----
C NTFS Fixed Healthy OK 73.59 MB 559 MB
D NTFS Fixed Healthy OK 387.28 GB 953.21 GB
E Unknown Removable Healthy Unknown 0 B 0 B

PS C:\Users\RODRIGOMENENDEZMOLIN>
```

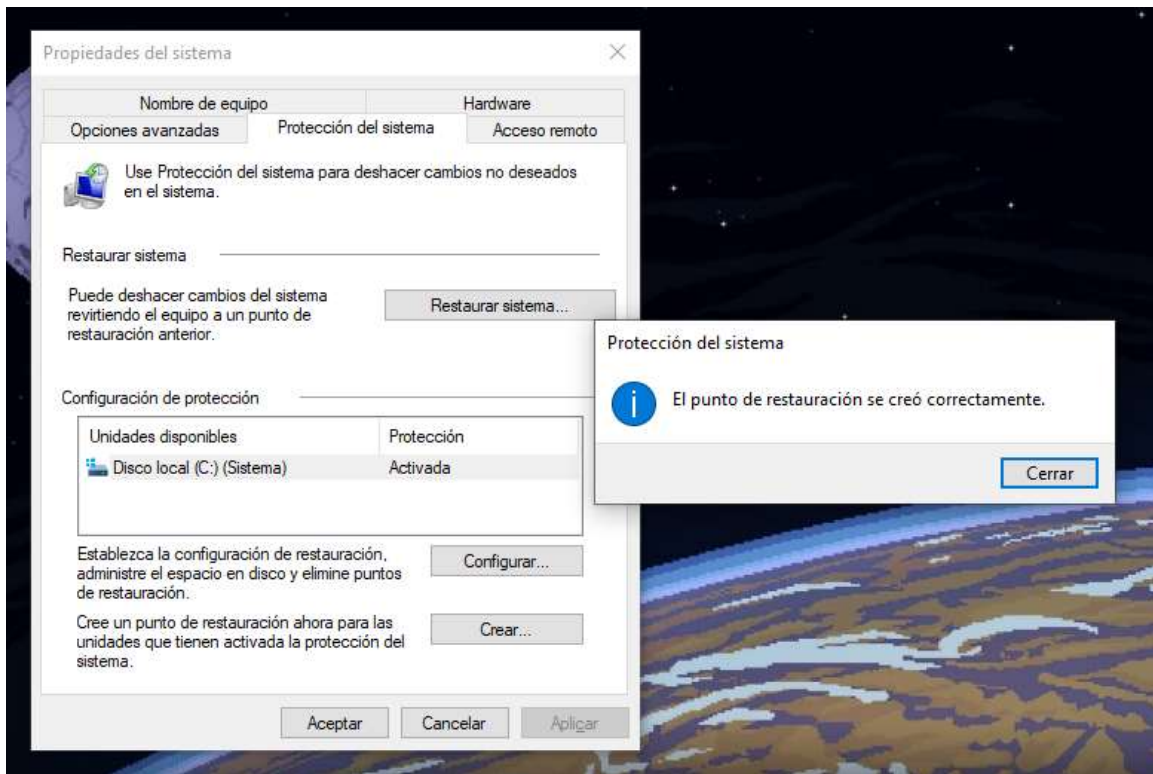
Práctica 2: Limpiar disco rápidamente

1. Win + R → escribe cleanmgr
2. Selecciona unidad C:
3. Marca:
 - Archivos temporales
 - Papelera de reciclaje
 - Archivos temporales de internet



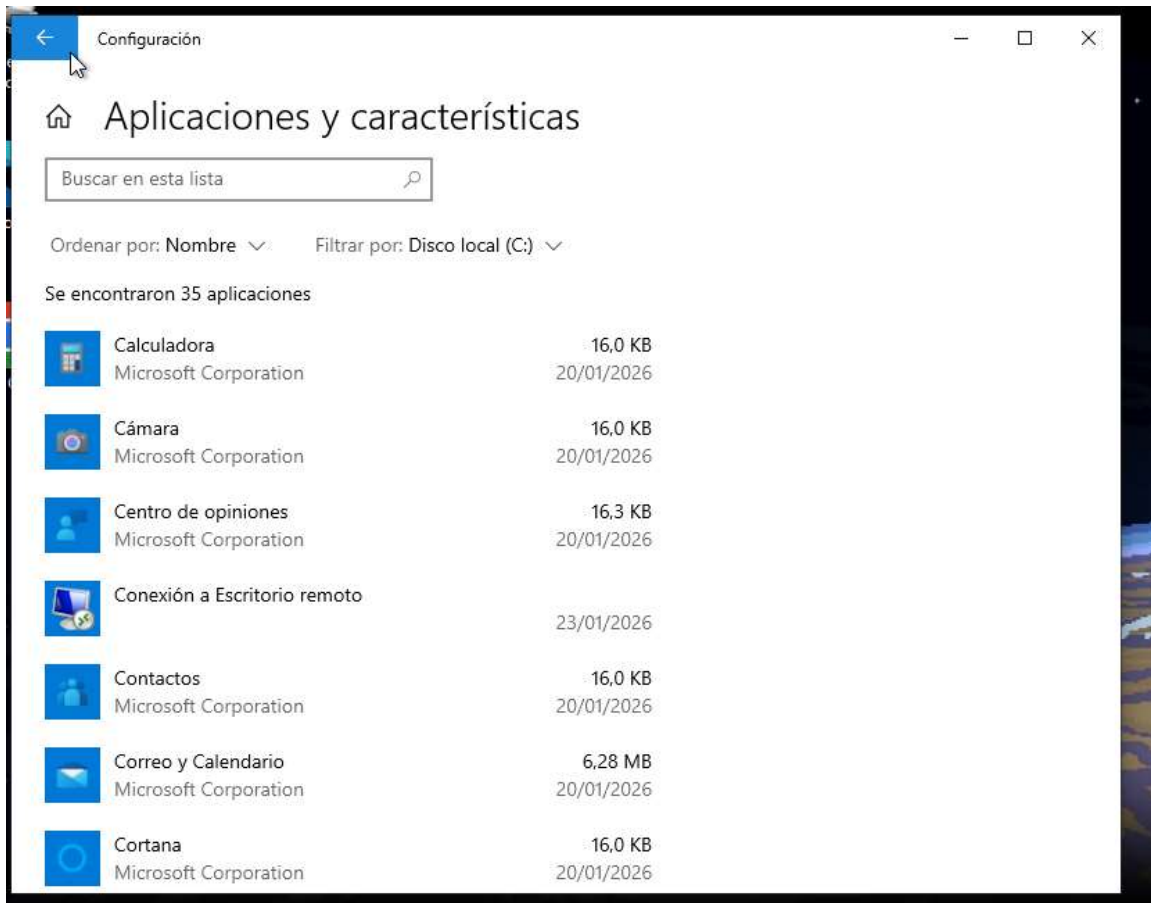
Práctica 4: Crear punto de restauración manual

1. Win + R → escribe sysdm.cpl
2. Ve a pestaña Protección del sistema
3. Selecciona unidad C: → Haz clic en Configurar
4. Activa 'Activar protección del sistema'
5. Regresa y haz clic en Crear
6. Pon nombre: 'Punto de práctica' → Crear



Práctica 5: Ver uso de disco por aplicación

1. Win + I (Abrir Configuración)
2. Ve a Sistema → Almacenamiento
3. Haz clic en Mostrar más categorías
4. Verás qué tipos de archivos ocupan espacio
5. Haz clic en Aplicaciones y características para ver por programa



PRÁCTICA EXTRA PARA AMBOS

Comparar resultados entre sistemas

1. En Linux: Anota cuánto espacio libre tienes con `df -h`
2. En Windows: Anota cuánto espacio libre tienes con Get-Volume
3. Compara: ¿Qué sistema muestra mejor la información?
4. Calcula: ¿Cuánto % de tu disco está usado?